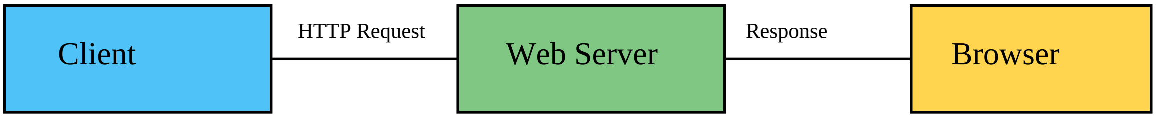
1. Explain the difference between Frontend, Backend and Full-Stack Development

Frontend: Builds the user interface using HTML, CSS and JavaScript.
Backend: Handles server logic, authentication and database.
Full-Stack: Develops both frontend and backend.
Example: In Amazon, product pages are frontend, order processing is backend, and a full-stack developer can build both.

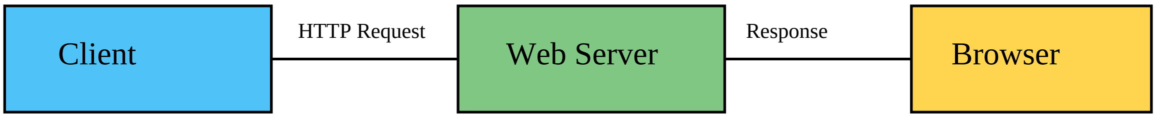
2. Client-Server Model

The client (browser) sends an HTTP request to the web server. The server processes the request and sends an HTTP response back.



3. Browser Requests a Web Page

Steps: Enter URL → Browser sends HTTP request → Server processes request → HTML/CSS/JS returned → Browser renders and displays the page.



4. Tools Required

VS Code (Code Editor), Browser (Testing), Live Server (Auto Refresh), Git (Version Control), GitHub (Cloud Repository), Node.js (JavaScript Runtime).

5. What is a Web Server?

A web server stores website files and delivers web pages to users over HTTP/HTTPS. Examples: Apache, Nginx, IIS and LiteSpeed.

6. Roles

Frontend Developer: UI.
Backend Developer: Server logic.
Database Administrator: Database security, backups and optimization.

7. VS Code Setup

Install VS Code and extensions: Live Server, HTML CSS Support and JavaScript snippets. Attach your own screenshot.

8. Static vs Dynamic Websites

Static websites show fixed content. Dynamic websites generate content using databases. Example: Portfolio (Static), Amazon (Dynamic).

9. Browsers

Chrome (Blink), Edge (Blink), Firefox (Gecko), Safari (WebKit), Opera (Blink). Rendering engines interpret and display web pages.

10. Basic Web Architecture

The client sends a request to the web server. The server communicates with the database and APIs, then returns the response.